

RV	Probability DISTR. / Density	$E[X]$	$\text{var}[X]$
Bernoulli:	$X_B(\omega) = \begin{cases} 1 & \text{"H"} \\ 0 & \text{"T"} \end{cases}$ $P(X) = \begin{cases} p & x=1 \\ 1-p & x=0 \\ 0 & \text{otherwise} \end{cases}$	p	$p(1-p)$
Binomial:	$X_B = \sum_{i=1}^N X_i$ $P(X_B=k) = \begin{cases} \binom{N}{k} p^k (1-p)^{N-k} & k \geq 0 \\ 0 & \text{otherwise} \end{cases}$	Np	$Np(1-p)$
Geometric:	$X_G = i$ s.t. X_i is the 1st "H" $P(X_G=k) = \begin{cases} (1-p)^{k-1} p & k \geq 1 \\ 0 & \text{otherwise} \end{cases}$	$\frac{1}{p}$	$\frac{1-p}{p^2}$
Poisson	$X_P = \# \text{ of events in unit interval}$ $P(X_P=k) = \begin{cases} \frac{e^{-\lambda} \lambda^k}{k!} & k \geq 0 \\ 0 & \text{otherwise} \end{cases}$	λ	λ
Uniform	$X_U = x \in [a, b]$ $p(x) = \begin{cases} \frac{1}{b-a} & x \in [a, b] \\ 0 & \text{otherwise} \end{cases}$	$\frac{a+b}{2}$	$\frac{(b-a)^2}{12}$
Exponential	$X_E = x$ in a Poisson Proc. time btw events $p(x) = \begin{cases} \lambda e^{-\lambda x} & x \geq 0 \\ 0 & \text{otherwise} \end{cases}$	$\frac{1}{\lambda}$	$\frac{1}{\lambda^2}$
Normal	$X_{Nor} = x$ on real axis $p(x) = \frac{e^{-\frac{(x-\mu)^2}{2\sigma^2}}}{\sqrt{2\pi} \sigma}$	μ	σ^2